

Classwork 03//Computational Thinking, Convolutional Neural Net**Student Name:** _____**Points:** 30**A Number:** _____**Due:** Feb 17, 2026**1 Log (1+x) Layer (10 Points)**

Provide a code-snippet of how would you design a neural network Layer using PyTorch by inheriting from `nn.Module` that performs $\log(1 + x)$ transformation on the feature matrix.

2 Convolution Dimension (5 Points)

Consider a kernel of dimension 3×3 operating on an input image dimension of 5×5 with a stride of 1. Consider no padding, what is the dimension of the output??

3 Convolution Computation (10 Points)

Given input matrix as

$$I = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 1 \\ 7 & 8 & 9 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

and kernel as

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Calculate the output feature map using convolution with a stride of 1.

4 Max Pooling

(5 Points)

Apply max pooling with a stride of 2 to

$$I = \begin{bmatrix} 3 & 2 & 5 & 1 \\ 4 & 1 & 6 & 2 \\ 1 & 7 & 3 & 4 \\ 2 & 5 & 1 & 3 \end{bmatrix}$$